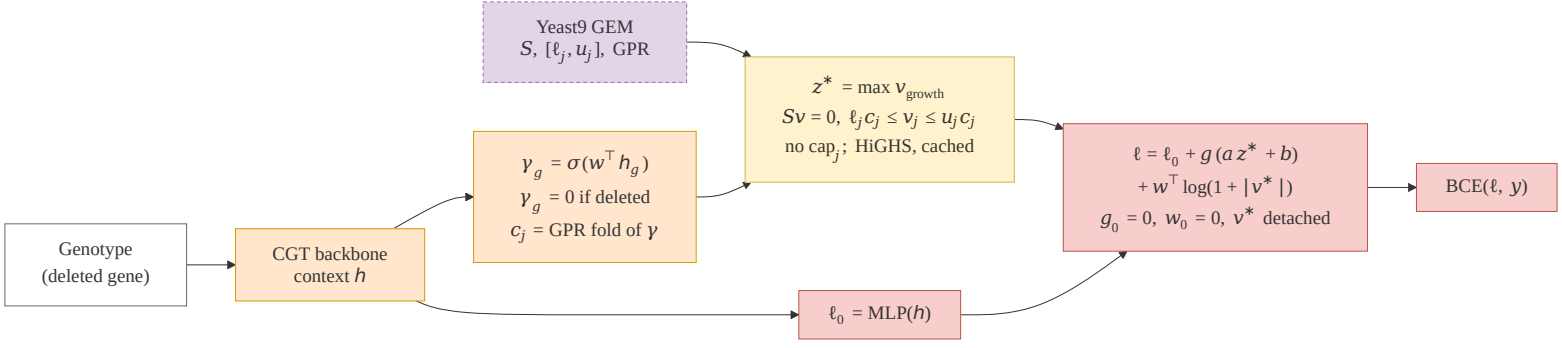

Yeast9 GEM
S, $[\ell_j, u_j]$, GPR
Genotype
(deleted gene)
CGT backbone
context h
$\gamma_g = \sigma(w^\top h_g)$
$\gamma_g = 0$ if deleted
c_j = GPR fold of γ
$z^* = \max v_{\text{growth}}$
$Sv = 0,\ \ell_j c_j \le v_j \le u_j c_j$
no cap$_j$; HiGHS, cached
ℓ_0 = MLP(h)
$\ell = \ell_0 + g\,(a\,z^* + b)$
$+ w^\top \log(1 + |v^*|)$
$g_0 = 0,\ w_0 = 0,\ v^*$ detached
BCE(ℓ, y)